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Source: *International Journal of American Linguistics*, Vol. 19, No. 3 (Jul., 1953), pp. 216-222
Published by: The University of Chicago Press
Stable URL: http://www.jstor.org/stable/1263010
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JICAQUE AS A HOKAN LANGUAGE

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1. Jicaque of Honduras in the past has been listed as an isolated language or classed with Paya-Lenca and Chibchan,[1] but we find unmistakable evidence that it is a Hokan language. In this paper we set forth structural and vocabulary correspondences to prove the relationship, along with an analysis of the place of Jicaque within the stock based upon cognate percentages. While the main facts of the linguistic situation stand out clearly, the limited material available for most of the languages prevents a full development of phonological and morphological details at this time.

The presence in Jicaque of numerous Hokan-Siouan correspondences was first noted by Greenberg while making a comparative survey in preparing a paper, Historical Linguistics and Unwritten Languages, for the International Symposium on Anthropology of the Wenner-Gren Foundation.[2] The subsequent further study of the problem by both authors was done under the Linguistic Divergence Project sponsored by the Columbia Council for Research in the Social Sciences.

2. The phonological equivalences in Hokan have been largely established by Edward Sapir's work.[3] The Jicaque agreements are generally obvious. A special point is that x^w and x have been replaced by k. Among some local dialects of Jicaque, r and n replace the l found elsewhere. Stem-finals have sometimes been changed, presumably by contraction with an added suffix. Because of the sparsity of Jicaque material no attempt has been made to distinguish the dialects. This accounts for variant forms sometimes given and for the fact that there are sometimes two or more distinct equivalents.

Jicaque and other forms in this paper have been phonemically interpreted and orthographically uniformized; any errors in the phonemic interpretation are surely not of a character that would affect the lexical comparisons.

3. The search for cognates is considerably hampered by the meagerness of the available Jicaque vocabularies and by their emphasis on words, such as plant and animal names, which have little utility for comparison at a remote time depth. And similar problems exist for most of the other Hokan languages with which Jicaque has to be compared.[4] The extent of the compara-

[1] See Harry Hoijer, Introduction p. 25, Linguistic Structures of Native America (Viking Fund Publication in Anthropology 6, 1946); J. Alden Mason, The languages of South American Indians p. 174 (Handbook of South American Indian, BAE-B 143, vol. 6, 1950.

[2] Subsequently published in *Anthropology Today*, 265–86 (Chicago, 1951).

[3] Especially in The Position of Yana in the Hokan Stock (UCPAAE 13.1–34, 1917); The Hokan and Coahuiltecan Languages (IJAL 1. 280–90, 1917–20); The Hokan Affinity of Subtiaba in Nicaragua AA27. 402–35, 491–527, 1925).

[4] The material on Jicaque is taken from two sources: Walter Lehmann, Zentral Amerika (Berlin 1920), pp. 654 ff; Edward Conzemius, The Jicaques of Honduras, IJAL 2.163 ff. (1921–23). The Chontal is mainly from Francisco Belmar, Lenguas Indigenas de Mexico (Mexico 1905), pp. 188–217. Other principal sources: A. M. Halpern,

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tive list should be judged with this in mind. We list more obvious similarities, with emphasis on cases showing relationships of Jicaque, Hokan-Coahuiltecan and Gulf languages.[5] Comparison with other language stocks belonging to Sapir's Hokan Siouan grouping, is made in a few instances where the agreement is particularly striking. It is convenient to include also some Hokan comparisons without Jicaque cognates, where we need them for the lexico-statistic counts given farther on.

-k adjective suffix, in Jicaque pek *white*, tek *black*, wek *good*, brek *many*; Yuman milk *black*, ahotk *good*, epilk *warm*; Pomo kedakedak *red*; Tonkawa makik *yellow*, maslak *white*.

First person: Jicaque na-p *I*, na- vestigial prefix *my*; Yuma nʸa *I*; Esselen ene *I*, niš- *my*; Karok na, ni- *I*, na- *me*, nani- *my*; Washo le *I*, lew *we*; Chimariko noʔut *I*; Yana ai-ʔni-ʒa *I*; Comecrudo na *I*; Coahuiltec na- *I*; Karankawa naayi *I*; Tunica -ni *I*.

Jicaque ku-p *we*; Chontal ki- *I*; Chumash ki- *we*, *our*; Tonkawa ka *I*; Pomo ke *we*; Karok kin- *us*; Yana -ni-gi *we*; Salinan ke *I*, ka *we*; Chitimacha -ki *I*, *me*.

Second person: Chontal i-ma *thou*, i-manki *you*; Seri me *thou*; Yuma ma *thou*; Salinan mo *thou*, -m- *thy*; Pomo ma *thou*, mi *thee*, mi- *thy*; Esselen nemi *thou*, meniš- *thy*; Yana aiʔnu-ma *thou*; Washo mi *thou*; Chimariko mamut, mi- *thou*, m- *thy*; Shasta mai *thou*; Achomawi-Atsugewi mi- *thy*; Karok im *thou*; Comecrudo em- *thou*; Coahuiltec mai- *thou*, *you*; Chitimacha him *thou*; Tunica maʔa *thou*.

Third person: Jicaque hu-p *he*; Chontal -u-pe- *he*; Chumash ho *that*; Salinan ho *that*; Chitimacha hus *he*, ha *this*, ho *these*; Pomo u *that*, *he* (distant).

Jicaque na *that*, nahac *there*; Yuma nʸaa- *that*, -nʸ stem of pronouns; Salinan na *that*; Yana na-gu- *therefore*; Comecrudo na-ni *he*, na- stem of pronouns; Tonkawa naʔa *thus*; Chitimacha naat *thus*; Coahuiltec namo *then*.

Jicaque kiwa *here*; Washo wa *here*, widi *this*; Yuma vaδi *here*, vaδa *this*; Chitimacha we *the*, *that*.

Interrogative: Jicaque kanlepa *when*, kat *where*; Chontal kana *when*, kaape *where*; Washo kunya *where*, kunyate *what*; Tunica ka- interrogative; Coahuiltec xat *how many*, xakat *why*; Comecrudo xat *where*, tokom *whatever*, *thing*; Chimariko qho- interrogative.

Yuma av- *what*; Yana ʔaci *what*, ʔambi *who*; Pomo am *who*, *what*; Chimariko ʔawilla *who*; Chitimacha ʔam *what*, ʔašt *how*, ʔampinki *where*.

Locative: Yuma -k *at*; Yana gi location, object relationship; Chitimacha ki *at*, kin *with*; Atakapa -ki *in*, *on*, -ik *with*.

one: Jicaque pani; Chumash paka; Tonkawa pax; Esselen pek; Yana bai; Chimariko pʔun; Coahuiltec pil; Comecrudo pek-weten (cp. elek-weten two). Also Yuki pawa, powik.

two: Chontal o-keši; Seri kaxkum; Yuma xavik; Chimariko χoku; Yana ʔux-; Shastan xokʷa, hoki; Pomo ko, xo; Esselen xulax.

three: Jicaque kont; Chontal a-fan-e; Esselen xulap; Pomo xoxat; Yuma xamok; Chitimacha kaayči; Karankawa kaxayi. Proto form χʷa.

five: see *hand*.

not: Jicaque kua *none*; Chontal k-; Chimariko χu-; Pomo kuyi; Yana ku-; Karankawa kom; Natchez koošac *no*; Tonkawa ooko *no*.

all: Jicaque put; Yana buiza-.

arm-hand: Jicaque pul, pel *arm*, pus

Yuma IJAL 12.25–33, 147–51, 204–12, 13. 18–30, 92–107, 147–66 (1946–7); Edward Sapir and Morris Swadesh, Yana Dictionary, ms. in American Philosophical Society Library; A. L. Kroeber, The Washo Language of East Central California and Nevada, UCPAAE 4.251–317 (1907); John R. Swanton, Linguistic Material from the Tribes of Southern Texas and Northeastern Mexico (BAE-B 127, 1940).

[5] Gulf is Mary R. Haas' term for 'the Muskogean family of languages plus Natchez, Tunica, Chitimacha and Atakapa.' See The Proto-Gulf Word for *Water* (IJAL 17.71–79, 1951).

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shoulder; Chumash pu *arm, hand*; Chimariko hitan-pu *arm*; Salinan puku *arm*; Subtiaba pahpu.

ashes: Jicaque pe; Chontal ƛ-api.

back: Yuma -mak; Yana maḱi.

bad: Jicaque marara; Yana malla-.

belly: Jicaque kol *belly*; Chontal ƛ-ikuu; Pomo koy, xo; Chumash akšu; Washo ciguguš; Cotoname kox; Atakapa kom.

big: see *old*.

blood: Jicaque kat, ac; Chontal awas (perhaps read awac); Seri avat; Yuma x^wat; Chumash axulis; Salinan ekata; Pomo hata; Yana wadʔdu-wi; Shasta-Achomawi axta; Karok ax; Comecrudo kyal; Coahuiltec k^was, hac; Subtiaba eʔdi. Proto form ax^wat varying with x^wat; Jicaque kat from x^wat, ac conceivably from ax^wat-s with weakening of intervocalic x^w but may be unrelated. See *red*.

bone: Jicaque kere; Chontal l-ekaƛ; Washo di-bigeñ; Natchez ek^wel; Creek funi (from x^wani); Comecrudo xi. Other languages show reflexes of x^wa(n)- in compound with other elements: Chitimacha kaci; Atakapa ci (from kci); Seri itak; Chimariko hitχun; Yuman tyaka; isaka; Chumash seʔ; Salinan axaak; Pomo ihya; Yana ixdal-la.

brother: Jicaque na-pey *brother-in-law*; Washo peyu *younger brother*; Chumash pe *older brother*.

child: Jicaque way; Seri ove.

chin, beard: Jicaque cek; Shasta cawak; Chumash acüs (*beard*), Yana ązawadći.

cold: Jicaque coys; Chontal ćoos; Yuman čuq, Kiliwi axčaq; Yana xaa-ćidʔ-; Chimariko xaca; Achomawi ašča; Tonkawa hacex; Atakapa akcaw; Alabama kasatka. Proto form perhaps $xacax^w$.

die: Jicaque te-puy; Yuma ʔapooy. Perhaps also cp. Tunica lupi; Chitimacha nuupi-.

drink: Jicaque mes; Atakapa em; Salinan ešim; Yana sii-; Yuman Mohave iθi; Seri k-asi; Karok is; Chontal šwa. Perhaps from imasi and isima as metathetized variants.

excrement: Jicaque piye; Shasta ipxai (*intestines*); Yana pa-ći; Salinan pxat; Chimariko h-ipxa (*intestines*); Subtiaba amba (from apa).

ear: Chontal išmac; Yuma ismalʸka; Chimariko hisam; Pomo šima; Yana malʔgu; Seri islo. Also Yuki cema, šum.

earth: Jicaque ma; Chontal ƛ-amac; Seri amt; Yuma amat; Chimariko ama; Pomo ama; Comecrudo kamla. Also Siouan ma.

eat: Jicaque te-k; Chontal -ka (*bite*); Salinan išax; Tonkawa yaxa-; Comecrudo kay; Chitimacha ḱaht- (*bite*), ḱušt- (*eat*).

eat: Yuma ama; Yana ma-; Chimariko hamma; Shastan -am-; Esselen am; Chumash umu; Salinan amo; Muskogi hump-.

egg: Jicaque pehey; Chontal ƛ-apik. Also Pawnee lipiku.

eye: Jicaque wut; Atakapa wul; Salinan ṭ-uwu (*face*).

fat, grease: Jicaque ponik; Salinan upikit; Chimariko -xu- (*obese*); Yana pui-; Achomawi aphau.

fire, hot, firewood: Jicaque awa *fire, hot*; Yuma aʔaw *fire*; Salinan t-aʔaw *fire*; Yana ʔau-na *fire*; Chimariko ʔaʔpu *fire*, h-au-na *tinder* (-na *wood*); Shasta awa *wood*; Subtiaba aaguu *fire*.

fly: Chontal -yu; Yuma ayer.

foot: Chontal imic; Yuma iime; Washo mayop; Comecrudo emi.

hand: Jicaque mas; Comecrudo mapi; Coahuiltec maw; Karankawa ecma (*hand*), ko-maso-pani (*five*); Chontal a-mahši; Yana ząman-. Also Keresan -amast-; Yuki mee, miipaʔt.

hand: Yana dal-la; Washo tatu; Yuma isalya-; Achomawi ili (from itli); Chimariko hican; Pomo tana. Chimariko hitan-pu *arm* and χuli-teni *left hand* seem to contain the same element but without the change of t to c; this can be explained if we assume two forms, itan and tan, deriving hican from hi-ican but hitanpu from hi-tan-pu, or by assuming variant initial consonants.

hear: Jicaque pakas; Yana ga-.

heart: Jicaque -has; -hos; Chontal husʔančey; Coahuiltec xasaal; Tonkawa yacanan; Seri imos; Washo d-emli.

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horn: Jicaque seme; Chitimacha ʔims; Comecrudo yemo; Cotoname yomo.

hot: Jicaque pe; Chontal epal; Yuma ʔapilʸ.

house: Jicaque wa; Yuma ʔava; Esselen iwa-no; Chimariko ʔawa; Yana wa-wi; Comecrudo wamak; Subtiaba gʷa.

kill: Jicaque ti-kwaw; Pomo kum; Chimariko -ko-; Comecrudo kamaw; Chitimacha ḱet-. Also Keresan kaut.

leg: Jicaque tek; Chontal itungu.

meat: Jicaque pisis; Pomo peše; Yana basi; Salinan ṕaš (*elk*); Chitimacha kipi; Muskogean nipu; Chontal l-išik.

man: Jicaque kuy; Chontal x̌-akuy; Diegueño ikuč; Pomo xak; Coahuiltec xaku; Tonkawa haʔakon; Chumash uxuy; Yuma nʸako (*father*); Chumash gogo (*father*). Also Yuki ḱew. Compare *old*; the word for *man* may come from the word for *adult*.

mother: Jicaque nee; Shastan ani; Pomo nixa; Yana nii-na; Washo -la.

mouth: Jicaque lam, lala; Tonkawa kala; Comecrudo xal; Atakapa kat; Yuma axaʔa; Pomo xacita; Chumash ïk; Chitimacha ša; Tunica šohu; Karankawa emi-akoy. Earliest form perhaps xa-la, with development in two directions as lala and axa, ixa.

near: Jicaque nya-pin; Yuma xiipan; Atakapa ipal; Chitimacha paye (*proximity*); Karankawa pawail.

night: Jicaque piste; Yana basii; Cotoname pay; Washo ta-pawit; Yana paawi-sa (*night falls*); Shasta upxa. Also Keresan ko-pso; Yuchi pato.

nose: Jicaque mik (perhaps from m-yik); Seri hif, wuf; Yuma iixuu; Esselen hošis; Chimariko hoxu; Achomawi yammi; Karok yufi; Washo šuyep; Comecrudo yax; Cotoname yaʔx; Atakapa uc (from yuk-s); Chitimacha šiš (assimilated from yiš); Tunica riši; Muskogi yupo; Subtiaba d-aako. Perhaps yaxu, with various vowel assimilations.

old: Jicaque kokoy; Chontal akuy (*old*), -koy (*big*); Seri kakox (*big*); Chumash no-χaχ (*big*); Chimariko xawini; Yana xaaba-; Yuma kʷoraʔaka (*old man*).

rain: Jicaque yipa (*rain*), hipe (*it is raining*), yupu (*water*); Seri ipka; Natchez yoba; Muskogean Alabama oyiba, Hitchiti ooba.

rain: Chontal ukuy; Yuma ikʷe-; Chitimacha kaya.

red: Jicaque ke; Pomo kedakedak; Atakapa kuc (from kut-s from ketu); Pomo kača; Yuman koxʷat, ečawata, xʷat; Yana ba-wadʔ-; Karok ačut; Shasta ikʷata; Achomawi ačte. Possibly the same element as in *blood* or a different element which has become intertwined in some of the languages.

river: Jicaque cew; Tonkawa cakaw; Chitimacha čaʔaał.

salt: Jicaque carin; Chitimacha ćahća (*salty*).

sing: Chontal šow; Yuma aašvaar.

sinew: Jicaque pupuy (*vein, nerve*) pey (*rope*); Yana baama; Shastan imme, pim, ippiu.

sky: Jicaque mol (*cloud*); Chontal l-emaa; Seri amime; Yuma ʔamay; Yana ząbząa; Salinan lem (*above, sky*).

sleep: Jicaque maha; Salinan me; Karankawa im; Comecrudo eme-t; Subtiaba ami; Chontal šmay; Seri sim; Yuma ašma; Pomo sima; Yana samsi-; Chimariko pohmu; Shastan icmas ici; Coahuiltec comoxʷam. Also Keresan sipa-.

smell: Jicaque kucuway; Chontal -we.

stone: Jicaque pe; Chontal apik; Yuma ʔavii; Chumash χïp; Pomo xape; Esselen šefe; Salinan šxap; Chimariko qhaʔa (*stone*), qhaku (*arrowpoint*); Yana xaga (*flint*); Pomo xaka (*knife, obsidian*); Salinan išak (*knife*), asaka (*flint*); Achomawi sat (*arrowpoint*). Evidently two elements, pe and qhak, which combine to form a compound in Pomo and some of the other languages.

sun: Jicaque laksak; Atakapa lak (*sun, glitter*); Pomo la (*sun*), alaša (*moon*); Comecrudo al; Muskogean Choctaw luwak (*fire*); Seri sax (*sun*), isax (*moon*); Salinan esxay (*morning, day*); Esselen asaca (*day*), asi (*sun*); Chimariko ʔasi; Shasta ačayi. The Jicaque is evidently a compound of

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two elements, perhaps ala-k *glowing* and sax *sun*.

tongue: Jicaque peraŋ; Chontal ƛ-ipaƛ; Yuma iipal^{y}; Seri ipl; Pomo pal; Shasta ipli; Chimariko h-ipen; Karok apri; Yana bawalʔ-la; Yana bal-la (*mouth*); Comecrudo expel; Atakapa nel; Chitimacha wen.

tree: Yuma ʔaʔii; Yana ʔi-na; Esselen iʔi.

wash: Jicaque te-pays; Chontal -pah; Chimariko -pok-; Comecrudo pelepla; Atakapa pac (from pak-s).

water: Chontal l-aha; Seri ax; Yuma axa; Pomo xa, ka; Chimariko aqha; Yana xa-na; Karok aš; Shastan aca, aš, acsi; Salinan ṭ-ša; Esselen asa-nax.

white: Jicaque pek; S. Chumash pupu; Chontal fuka; Karankawa peka; Comecrudo pepok; Washo tal-popoy.

wind: Chontal ƛ-awa; Washo t-ewesgiw.

woman: Jicaque kep, kek (*woman*), nakom (*wife*); Chitimacha kiča; Atakapa kis; Comecrudo kem; Seri kmam; Cotoname katam; Tonkawa k^{w}aan; Chimariko punsar (from k^{w}em-sa-r).

woman: Jicaque mumuy (*female*); Yana mariʔmi; Washo tamomo (*woman*), umlaya (*wife*); Pomo mata; Comecrudo mat (*mother*).

4. That Jicaque is related to Hokan-Coahuiltecan is clearly evident from the quality and quantity of agreements. The next problem is to determine its exact position in the stock. The most convenient objective method for doing this is that of lexical statistics, based on percentages of cognates in a test vocabulary.[6] Using a test list of 200 items, belonging to the sphere of every-day and relatively non-cultural meanings, we match them in each of the languages insofar as our material permits. The number of cognates is considered by reference to the total number of pairs available for comparison.

[6] See Swadesh, Lexico-Statistic Dating of Prehistoric Ethnic Contacts, APS-P 96.452–63 (1952).

We give the results of cognate counts for six languages—Jicaque, Chontal, Yuma, Yana, Washo, Comecrudo—taken as a fairly representative selection of the whole stock. Yana is taken as one of the Northern Hokan languages, a substock which also includes Pomo, Chimariko, Karok and Shasta-Achomawi. Yuma is taken as a representative of Seri-Yuman, because the available vocabulary is a little larger than Seri and it was found that both languages give approximately the same cognate percentages. Esselen, Chumash and Salinan were omitted because of the limited vocabularies available. Only one of the Coahuiltecan languages is included, namely Comecrudo, which has the advantage of showing higher percentages of agreement with the Hokan languages than some of the others. We thus avoid the uncertainties of dealing with minimal percentages, approaching the quantitative area of pure chance.

Because of the limited lexical materials available for Jicaque and some of the other languages, we have available in the different pairs of languages as little as 55 pairs in some cases and never more than 89 pairs. It will some day be possible to improve this situation. We are also handicapped by the fact that, even with Sapir's several papers, the study of etymologies is not yet sufficiently developed to guarantee accurate identification of cognates in many cases. In deciding what to include and what to exclude, we believe we have erred somewhat more on the side of overcaution than otherwise.

5. Following the method outlined, we obtained the results indicated below. For each pair of languages, we give the English for the items counted as cognates. The actual forms can be found in the cognate list above (**3**). To conserve space we do not list the items scored as non-cognates (e.g. Jicaque laksak, Chontal piču *day* or Jicaque pisis, Chontal lišik *meat*).

For Jicaque-Chontal, there were 79 items

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of meaning which could be matched in both languages. Of these, we apparently have cognate forms for identical meanings in the following 19 cases: *three, ash, belly, blood, bone, cold, earth, eat, egg, leg, man, old, sleep, smell, stone, tongue, when, where, white.*

For Jicaque-Yuma, there are 17 cognates in 73 pairs: *I, here, that, when, three, blood, die, dust, earth, eye, fire, hot, nose, sleep, stone, tongue, woman.*

For Jicaque-Yana, 12 cognates in 74 pairs: *I, one, three, all, bad, fire, hear, meat, night, old, rain, tongue.*

For Jicaque-Washo, 11 cognates in 75 pairs: *I, here, where, belly, bone, drink, green, head, mother, night, white.*

For Jicaque-Comecrudo, 13 cognates in 89 pairs: *one, blood, bone, earth, hand, husband, kill, mother, sleep, tongue, wash, white, women.*

For Chontal-Yuma, 11 cognates in 48 pairs: *blood, ear, fly, foot, hot, rain, sky, stone, tongue, water, sing.*

For Chontal Yana, 13 cognates in 75 pairs: *thou, two, five, blood, ear, hear, meat, not, salt, sleep, this, tongue, water.*

For Chontal Washo, 7 cognates in 55 pairs: *thou, you, two, belly, bone, where, wind.*

For Chontal-Comecrudo, 15 cognates in 65 pairs: *I, we, bird, bone, earth, fall, fish, foot, hear, mouth, not, sleep, tongue, water, white.*

For Yuma-Yana, 15 cognates in 67 pairs: *what, at, two, three, back, blood, drink, ear, eat, fire, hand, sleep, tongue, tree, water.*

For Yuma-Washo, 7 cognates in 55 pairs: *I, here, this, eat, father, foot, nose.*

For Yuma-Comecrudo, 7 cognates in 56 pairs: *blood, earth, foot, nose, sleep, tongue, water.*

For Yana-Washo, 11 cognates in 92 pairs: *I, thou, we, two, dog, eat, hand, mother, night, say, woman.*

For Yana-Comecrudo, 12 cognates in 84 pairs: *one, blood, blow, fall, liver, not, rain, skin, sleep, snow, tongue, water.*

For Washo-Comecrudo, 6 cognates in 61 pairs: *all, bone, foot, mouth, nose, white.*

6. Reducing the foregoing figures to percentages, we find the following relationships:

	Jicaque	Co	Wa	Ya	Yu
Chontal.....	24	23	13	19	23
Yuma.....	23	13	13	21	
Yana.....	16	14	12		
Washo.....	15	10			
Comecrudo....	15				

Presumably the different degrees of relationship are based on the extent of contact among the Hokan languages when they were first diverging, that is, when they were regional dialects of a single language. The cognate percentages therefore suggest an approximate relative geography of early Hokan.

The lowest cognate percentages among these languages, 10%, when interpreted by the lexico-statistic constant of 66% (±4%) common retention after 1000 years, indicate that they represent the differentiated products of what was essentially a single language approximately five thousand years ago (calculated as 55 ± 7 centuries). Within a few centuries of that time, local dialects must have begun to show differences. Presumably contiguous dialects were largely the same while non-neighboring dialects were already noticeably divergent. Presumably, at that time, Comecrudo's neighbors to the east, the dialect forerunners of the present day Coahuiltecan and Gulf languages, shaded off in the same general way with lexical differences approximately proportional to the relative place of the dialects in the chain. Comecrudo fits into the dialect network of Hokan, since it is not more different from Chontal than the latter is from Jicaque or Yuma. This suggests that there is no sharp separation between Hokan and Coahuiltecan, that Hokan and Coahuiltecan and presumably

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also Gulf are not discrete stocks or branches but merely three segments of the same continuous chain.

7. The present analysis helps to clarify the internal relationships of Hokan-Coahuiltecan to some extent, but much remains to be done both in verifying the conclusions reached so far and in extending the analysis to the other languages in the Hokan-Coahuiltecan-Gulf complex. To facilitate such study, it is clear that we need more complete vocabularies as well as more adequate phonetic and structural analyses. It is also necessary to exploit to the full the possibilities of reconstructing phonology and of checking etymologies afforded by the closer subgroupings within the total complex. For example, the study of Shasta-Achomawi-Atsugewi or of Seri-Yuman would make it possible to take elements back a thousand or two years in time, filling out the details of the earlier phonetic form, and would therefore greatly increase the assurance with which we can compare words of totally different subdivisions of Hokan.

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